MIL-S-7809A
3 MAY 1963
SUPERSEDING
MIL-S-7809
28 NOVEMBER 1951

MILITARY SPECIFICATION

STEEL, HIGH-STRENGTH, LOW-ALLOY, BARS, SHAPES, SHEET, STRIP, AND PLATE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers high-strength low-alloy hot rolled steel bars, shapes, plates, sheet, and strip products intended primarily for use in welded structures.

1.2 Classification. Materials shall be furnished in the following forms and sizes, as specified (see 6.2).

1.2.1 *Forms.*

Bars
Plates
Sheet
Strip — cut lengths and coils
Shapes
Bar-sized shapes

1.2.2 *Sizes.*

(a) Less than ½ inch in thickness
(b) ½ to 1½ inches in thickness
(c) Over 1½ to 4 inches in thickness

1.2.3 *Composition.* Low-alloy high-strength steels shall conform to one of the chemical compositions of table I (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-6868 — Inspection Process, Magnetic Particle

MIL-E-15599 — Electrodes, Welding, Covered — Low — and Medium-Carbon Steel

STANDARDS

FEDERAL

FED. STD. No. 48 — Tolerances for Steel and Iron Wrought Products

FED TEST METHOD STD NO. 151 — Metals; Test Methods

MILITARY

MIL-STD-163 — Preparation of Steel Products for Do-

FSC 9510, 9515, 9520

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mestic Shipment (Storage) and Overseas Shipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATION

A 233-58T — Tentative Specification for Mild Steel Arc-Welding Electrodes

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 **Material.** The steel shall be made by the basic oxygen, open-hearth or electric furnace process, and shall be furnished in the as-rolled, annealed, or normalized condition.

3.2 **Chemical composition.** Ladle analysis shall conform to the limits as specified by table I for the specified grade (see 6.2).

3.2.1 The manufacturer shall furnish a ladle analysis of each heat of steel from which material is furnished on the contract. When steel composition is specified and check analyses required or the materials are not identifiable by heat or melt, individual determinations shall be within the respective table I analysis as modified by the applicable check analysis tolerance.

3.3 **Mechanical properties.** The mechanical properties shall be as specified in table II.

3.4 **Bending.** Transverse bend test specimens shall withstand being bent at room temperature through 180 degrees without cracking on the outside of the bent portion. The inside diameter of the bend shall have such relationship to the thickness of the specimen as prescribed (see 4.5.2).

3.5 **Weldability.** All steel furnished under this specification shall be capable of being fusion welded (see 4.6.2), without preheating, at normal room temperature of 60° F or higher, without developing cracks in the heat-affected zone adjacent to the deposited weld metal.

3.6 **Finish.** Hot rolled products will normally be furnished as rolled. When specified, the scale or oxides shall be removed by pickling or shot blasting and shall be coated with oil. The as-hot-rolled steel shall be furnished dry. Cold-rolled products are normally furnished as rolled.

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TABLE I. *Chemical composition* [1]
(percent)

Grade	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Chromium	Nickel [2]	Copper [2]	Molybdenum	Zirconium	Aluminum	Vanadium
945A	0.15	1.00	0.040	0.050	0.90	—	—	0.30 min	—	—	—	—
945C	0.23	1.40	0.040	0.050	0.90	—	0.50–1.00	0.30–0.60	—	—	—	—
950A	0.15	1.30	0.040	0.050	0.90	0.40–0.65	—	—	—	0.05–0.15	—	—
950B	0.23	1.30	0.040	0.050	0.90	—	—	0.20–0.50	—	—	—	0.02 min
950C	0.25	1.60	0.050	0.040	0.90	—	—	—	—	—	—	—
950D	0.15	1.00	0.150	0.050	0.90	—	—	—	—	—	—	—
Composition 1	0.12	1.00	0.12	0.050	0.050	0.40–1.00	0.25–0.75	0.30–0.70	—	—	—	—
Composition 2	0.15	1.00	0.150	0.050	0.17	—	0.40–0.85	0.75–1.35	0.05–0.20	—	0.10–0.30	—
Check Analysis Tolerance %	0.03	0.05	0.01	0.01	0.05	0.03	0.03	—	0.01	—	0.05	0.02

[1] Maximum limits unless otherwise specified in the table.

[2] Except as otherwise indicated, copper and nickel shall not exceed 0.15 percent each, and molybdenum not exceeding 0.05 will be considered as residual and will not be cause for rejection of the melt.

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TABLE II. *Mechanical properties*

Product	Property	Grade 945	Grade 950 and compositions 1 and 2
Cold rolled sheets.	Yield strength, psi [1]	45,000	50,000
	Tensile strength, psi	60,000	70,000
	Elongation in 2 inches, %	20	20
Hot rolled sheets.	Yield strength, psi [1]	45,000	50,000
	Tensile strength, psi	60,000	70,000
	Elongation in 2 inches, %	22	22
Plates, shapes, and bars to ½ inch inclusive.	Yield strength, psi [1]	45,000	50,000
	Tensile strength, psi	65,000	70,000
	Elongation in 2 inches, %	22	22
Plates, shapes and bars over ½ to 1½ inclusive.	Yield strength, psi [1]	42,000	45,000
	Tensile strength, psi	62,000	67,000
	Elongation in 2 inches, %	24	24
Plates, shapes, and bars over ½ to 3 inches.	Yield strength, psi [1]	40,000	42,000
	Tensile strength, psi	62,000	63,000
	Elongation in 2 inches, %	24	24

[1] Yield strength is to be measured at 0.2% offset.

3.7 Conditioning of shapes, plates, and bars.

3.7.1 Structural-size shapes may be conditioned by the manufacturer for the removal of injurious surface imperfections or depressions on their surfaces by grinding, or chipping and grinding, provided the area is well ground and the depression does not extend below the rolled surface of the shape more than the following:

1/32 inch in material less than 3/8 inch in thickness, or

1/16 inch in material 3/8 inch and over in thickness.

Surface imperfections which are greater in depth than the limits shown above, may be removed by chipping or grinding and then depositing weld metal, subject to the following limiting conditions:

(a) The total area of the chipped or ground surface of any piece prior to welding shall not exceed 2 percent of the total surface area of that piece.

(b) After removal of any imperfections preparatory to welding, the thickness of the material at any location must not be reduced by more than 20 percent of the nominal thickness of the section, nor can this reduction in thickness ever exceed ½ inch in depth, at the location of the imperfection.

(c) The toes of angles, beams, channels, and zees and the stems and toes of tees may be conditioned by grinding or chipping and welding, provided the maximum depth of depression prior to welding does not exceed ½ inch.

3.7.2 Imperfections that do not affect the utility of the plates are not commonly considered injurious defects. Plates are commonly conditioned by the producer for the removal of surface imperfections or depressions on either surface by grinding, provided the ground area is well flared and grinding does not reduce the thickness of the as-rolled plate by the following values:

(a) More than 7 percent of the nominal thickness for plates specified to weight per square foot, but in no case more than 1/8 inch.

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(b) Below the minimum thickness for plates specified to thickness in inches.

3.7.2.1 Plates may have surface imperfections removed by chipping and then depositing weld metal, in the following manner:

(a) The chipped area of each surface of a plate shall not exceed 2 percent of the area of that surface.

(b) After removal of any imperfections preparatory to welding, the thickness of the plate at any location shall not be reduced by more than 20 percent of the nominal thickness of the plate.

(c) All welding shall be performed by welders using low hydrogen welding electrodes conforming to the proper series of the EXX15, EXX16, or EXX18 classifications of ASTM A233, unless otherwise specified. The electrodes should be protected from moisture pick-up during storage and use. The welds shall be sound and the weld metal shall be thoroughly fused to all surfaces and edges without undercutting or overlap. Weld metal shall project at least 1/16 inch above the rolled surface after welding, and the projecting metal shall be removed by grinding, or chipping and grinding to make it flush with the rolled surface and produce a workmanlike finish.

3.7.3 Bars may be conditioned by the manufacturer for the removal of surface imperfections by grinding, chipping, or some other means provided the conditioned area is well flared and the effective sectional area is not reduced by more than the permissible variations specified in table III.

3.8 **Identification marking.** All steel except bars, bar-size sections, and strip in coils or cut lengths shall be marked at least once near either end with the following information:

Specification number
Heat number
Grade

3.8.1 *Rounds, square, and hexagons.* The heat number or symbol shall be stamped on one end of each bar for sizes 1 inch and over in diameter or thickness flats. For smaller sizes than 1 inch, the information shall be shown on tags attached to the lift.

3.8.2 *Flats.* The heat number or symbol shall be stamped on one end of each bar for sizes 1 inch and over in thickness and width. For sizes under 1 inch in thickness and width, and information shall be shown on tags attached to the lift.

3.8.3 *Strip.* The information required in 3.8 shall be shown on tags attached to each coil or each lift for strip in cut lengths.

TABLE III. *Dimensional tolerances*

Form	Dimensional tolerance	Fed. Std. No. 48 paragraph references
Bars and bar-size sections [1]		
Rounds	Size and diameter, out-of-round	1a1, 1b1 [2]
Squares	Size and width, out-of-square	1a1, 1b1 [2]
Hexagons	Size and out-of-hexagon	1a2
Flats	Thickness and width	1a3
Bar-size angles	Thickness, length, out-of-square	1a4
Bar-size channels	Thickness of web, out-of-square	1a5
Bar-size tees	Width, thickness, stem out-of-square	1a6

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TABLE III. *Dimensional tolerances*—Continued

Form	Dimensional tolerance	Fed. Std. No. 48 paragraph references
Other bar-size sections	All dimensional tolerances	1a7
All	Random lengths	1a8
All	Specific lengths	1a9, 1a10
All	Length: Sheared, hot sawed	1a11
All	Straightness	1a12
All	Special straightened	1b6
Plates		
Sheared	Thickness and weight, when ordered to thickness	4a1
	Weight, when ordered to weight	4a2
	Thickness, when over 2 inches thick	4a3
	Width and length	4a4, 4b3 [2]
	Camber	4a9
	Flatness	4a11, 4b11 [2]
Universal mill	Thickness and weight, ordered to thickness	4a1
	Weight, when ordered to weight	4a2
	Thickness, when over 2 inches thick	4a3
	Length	4a4, 4b3 [2]
	Rolled width	4a5, 4b4 [2]
	Camber	4a9, 4a10
	Flatness	4a11, 4b11 [2]
Standard structural shapes		
Beams	All dimensional tolerances	9a2
H-beams	All dimensional tolerances	9a2
Channels	All dimensional tolerances	9a2
Zees	All dimensional tolerances	9a2
All	Weight	9a1
All	Length and ends-out-of-square	9a5
All	Camber	9a7
Sheets [3]		
All	Thickness [4]	11a3, 11b1 [2]
Cut edge	Width	11a4, 11b2 [2]
Mill edge	Width	11a5, 11b2 [2]
Not resquared	Length	11a6, 11b3 [2]
	Camber	11a12, 11b7 [2]
Cut edge, not resquared	Out-of-square	11a9, 11c5 [5]
Resquared	All dimensional tolerance	11a10, 11c6 [5]
Stretcher leveled [6]	Width and length	11a7, 11a8
	Flatness [7]	11a14,
Not stretcher-leveled [6]	Flatness [8]	11a13,
Strip		
Coils [9] and cut length	Thickness	14a1, 14b1 [2]
All	Crown	14a2, 14b2 [2]
All	Width	14a3, 14b3 [2]
All	Length	14a4, 14b4 [2]
All	Camber	14a5,
All	Flatness	14a6,

[1] A channel, angle, tee, or zee is in "bar" size when the greatest cross-sectional dimension is less than 3 inches.
[2] Applicable to class C alloy steels.
[3] All material shall be straight and flat within the indicated tolerances, except material ordered in coils which shall be free from buckles rendering the material unfit for the purpose intended.
[4] Tolerances shall be 0.001 inch greater than the tolerances shown herein.
[5] Applicable to class B high-strength low-alloy steels.

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[6] Refers to stretcher leveled standard of flatness.

[7] Such tolerances for sheets less than 0.0284 inch thick must be negotiated between purchaser and contractor.

[8] Allow an additional ¼ inch for permissible maximum height of buckles specified herein. However, for all steels, not more than two buckles of the maximum height shall be permitted on a side or in the middle of any sheet in any 120 inches of length. If more than two buckles are present, the height shall be proportionally lower, but an excessive number of buckles approaching corrugation shall be cause for rejection of the sheet. For sheet over 60 inches wide in 16 gage and heavier and over 36 inches wide, 17 gage and heavier, increase permissible maximum by 50 percent.

[9] Applicable only to high-strength low-alloy and high-strength alloy steel.

3.9 Dimensions and tolerances. The form and dimensions shall be as specified. Tolerances shall conform to Federal Standard No. 48 as specified in table III.

3.10 Workmanship. Within the limits of good manufacturing practice, the surfaces of all products shall be free from pipe, seams, laminations, blisters, excessive or detachable scale and shall not contain any other defects which, due to their nature, degree, or extent, prevent the fulfillment of other applicable requirements of this specification. Spot grinding to remove defects from sheet or plate is permissible provided minimum thickness tolerances are not exceeded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling and inspection for dimensions, finish, workmanship, packaging and identification marking requirements. Units of product shall be randomly selected to represent each lot of material of one heat, nominal dimensions, and offered for acceptance at one time, in accordance with table IV.

TABLE IV. *Inspection samples*

Lot size	Sample size	Acceptance number
1 to 7	all	0
8 to 40	7	0
41 to 110	15	1
111 to 180	25	2
181 to 301	35	3
over 301	50	4

4.2.1 *Method.* Samples shall be visually inspected.

4.3 Chemical analysis.

4.3.1 *Sampling.* When materials are not identifiable as to heat, samples shall be selected in accordance with table IV for check chemical analysis.

4.3.2 *Method.* Specimens shall be prepared in accordance with Federal Test Method Standard No. 151 and shall be tested by wet chemical, spectrographic, or other analytical methods. In the event of dispute, analysis shall be by wet chemical methods.

4.4 Tensile test.

4.4.1 *Sampling.* Two test samples shall be selected from each heat. If material from one heat differs ⅜ inch or more in thickness, one test sample shall be made from both the thickest and the thinnest material regardless of the weight represented.

4.4.2 *Method.* Test specimens or the samples from which they are machined shall be stress relieved before testing. The test specimen type shall be selected according to Method 211 of Federal Test Method Stand-

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ard No. 151 in accordance with table V. The location of tensile test specimens shall be as shown in table VI.

4.5 Cold bend test.

4.5.1 *Sampling.* Two specimens shall be selected from each heat. If material from one heat differs by 3/8 inch or more in thickness, one specimen shall be taken from both the thickest and thinnest material, regardless of the weight represented.

4.5.2 *Method.* Transverse bend test specimens shall be tested in accordance with Method 231 of Federal Test Method Standard No. 151. Specimens shall be bent cold through an angle of 180 degrees, over a diameter equal to the bend ratio times the thickness of the plate, sheet, or strip in accordance with table VII.

4.6 Weldability test.

4.6.1 *Sampling.* One sample shall be selected to represent each lot of steel submitted for inspection at one time, from one heat, of the same condition, and when heat treated, of the same process and batch. The sample shall be of sufficient size to prepare the weldability specimen shown in figure 1. The sample shall be of the maximum thickness found in the lot.

TABLE V. *Selection of test specimens for tensile tests from Fed. Test Method Std. No. 151*

Product or form	Product size (inches thick)	Fed Test Method Std. No. 151 Method 211 specimen	Comment
Bars, round	Under 1.50	Largest obtainable round specimen (R2, R3, R4, R5)	When practicable, test round bars in full section at option of contractor
Bars, flat	1.50 and over Under 1.50 1.50 and over	Largest obtainable round specimen (R1) F1 [1] or F2 [1] or R1 R1	
Plates, shapes (incl. bar-sized Shapes) sheets or strips	Under 0.250	F1 [1] or F2 [1]	
Plates and shapes (incl. bar-sized shapes)	0.250 to 1.0	F1 [1]	If impracticable, use specimen R1 or F2 at option of contractor
Plates	Over 1.0 to 2.0	R1 or F1 [1]	
Plates	Over 2.0	R1	

[1] Option: F3 may be used in place of F1 or F2.

TABLE VI. *Location of tensile test specimens*

Product of form	Product size (inches)	Location of longitudinal axis of test specimen
Bars, round	Under 1.50 diameter	Coincides with the central longitudinal axis of the piece
Bars, round	1.50 diameter and over	Coincides with the axis located midway between the surface and the center of the section
Bars, flat	Under 1.5 wide	Coincides with the central longitudinal axis

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TABLE VI. *Location of tensile test specimens*—Continued

Product of form	Product size (inches)	Location of longitudinal axis of test specimen
Bars, flat	1½ wide and over	Coincides with the axis located midway between the edge and center of the section
Plates, shapes, sheets, or strips	0.250	Coincides with the central longitudinal axis, or with the axis located midway between the flat surfaces, as applicable
Plates and shapes	0.250 to 1.0	Coincides with the longitudinal axis from a random position from a side edge or from a random position between a side edge midway between the center and edge of the plate with the axis of the specimen midway between the two flat surfaces, as applicable
Plates	Over 1.0 to 2.0	Random location at a side edge or random location between a side edge and center of the plate with the axis of the specimen midway between the center and surface
Plates	Over 2.0	
Bar, size shapes	All sizes	Full section or prepare test specimens 1½ inches wide (or smaller, if necessary)

TABLE VII. *Relationship of bend test inside diameter to thickness of specimen*

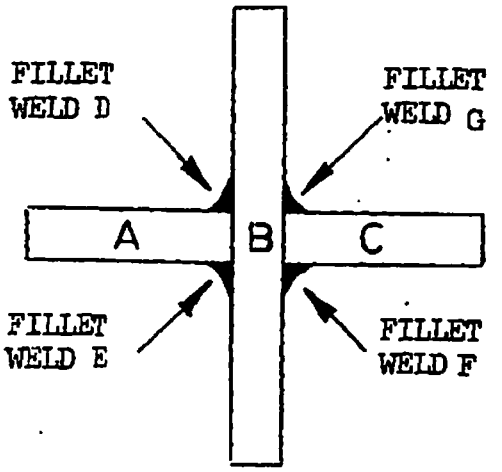
Thickness of material (inches)	Ratio of bend diameter to thickness of specimen
0.500 and less	1
0.501 to 1.500 inclusive	2
1.501 to 4.000 inclusive	3

4.6.2 *Welding procedure.* The size of the fillet welds on test specimens shall not exceed ½ of the thickness of test plates ⅜ inch and over (T/2 in figure 1). For test plate thickness less than ⅜ inch, the size of the fillet welds shall not exceed the plate thickness. Welds shall be made in one pass, except that fillet welds larger than ⅜ inch may be made in two passes. The initial temperature during welding for test specimens 1 inch or less in diameter shall not exceed 100° F. Test specimens over 1 inch thick may be preheated, at the option of the contractor. Specimens shall be permitted to cool to not less than 100° F nor more than 125° F between passes or between welds D, E, F, and G (see figure 1). A class MIL–6013 welding electrode conforming to Specification MIL–E–15599 shall be used to determine weld-crack susceptibility of the steel.

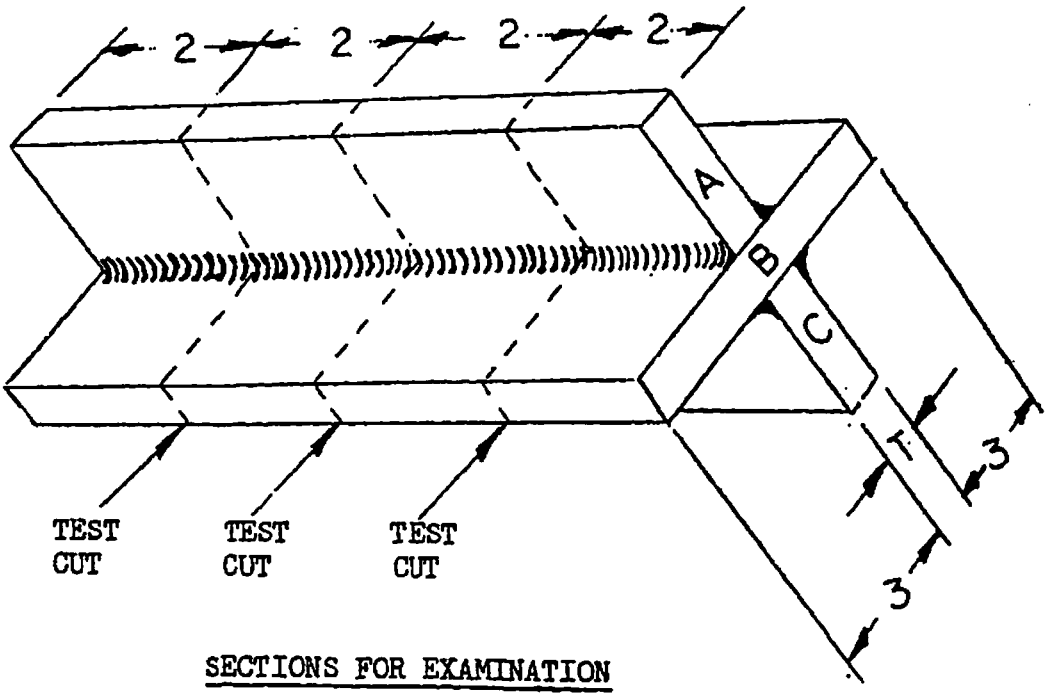
4.6.3 *Preparing specimens for examination.* After completion of test fillet welds, the specimens shall be allowed to remain in still air at room temperature for 24 hours before examination. Before sectioning for test cuts, the test specimen shall be visually examined at not more than 10 diameters magnification for cracks in the base metal. Following this examination, at least three test cut sections shall be taken not less than 2 inches apart and at least 2 inches from either end of the specimen, as shown in figure 1. Specimens may be annealed after welding and prior to sectioning in order to facilitate machining.

4.6.4 *Examination for cracks.* The test cut sections selected shall be surface-ground and examined with the aid of magnetic-particle inspection methods for cracks in the test plate adjacent to the test welds. The presence of cracks in the base metal shall be cause for rejection of the particular heat of steel that is represented by the test plate. The occurrence of cracks in the weld metal only shall not be cause for rejection. Specimens shall be examined for cracks adjacent to all welds in accordance with Specification MIL–I–6868. Examination shall be by means of the continuous method, using half-wave direct cur-

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ASSEMBLY OF SPECIMEN



SECTIONS FOR EXAMINATION

MACHINED FINISH SURFACES WHERE A AND C JOIN B.
DIMENSIONS IN INCHES (APPROXIMATE).
T = THICKNESS OF PLATE.

FIGURE 1. *Weld-crack-susceptibility test specimen (see 4.6)*

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rent magnetization and dry magnetic particles. A magnetization current of not less than 200 amperes nor more than 300 amperes shall be used.

4.7 **Packaging, packing, and marking.** Preparation for delivery shall be examined for conformance with section 5.

4.8 **Rejection.**

4.8.1 *Examination.* The lot shall be rejected if the samples selected fail to conform to the visual and dimensional requirements when examined in accordance with 4.2.

4.8.2 *Tests.* The lot shall be rejected for failure of the specimens to pass the tests specified in 4.3 through 4.6.

4.9 **Reinspection.**

4.9.1 *Reexamination.* Upon approval of the procuring activity, a lot rejected upon the basis of visual and dimensional examination may be resubmitted for examination providing the contractor removes all nonconforming material from the lot.

4.9.2 *Retests.* Retests shall be permitted in accordance with Federal Test Method Standard No. 151.

4.9.3 *Weldability.* Four additional specimens shall be prepared and tested in accordance with 4.6. All specimens shall meet the weldability requirements or the lot shall be rejected without further retesting.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.**

5.1.1 *Level A.* Preservation and packaging of steel shall be in accordance with Standard MIL-STD-163.

5.1.2 *Level C.* Preservation and packaging shall be in accordance with standard commercial practice.

5.2 **Packing.**

5.2.1 *Level A.* Packing of steel shall be in accordance with Standard MIL-STD-163.

5.2.2 *Level C.* Packing shall be in accordance with commercial practice adequate to ensure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 **Marking for shipment.** Marking for shipment shall be in accordance with Standard MIL-STD-163.

6. NOTES

6.1 **Intended use.** The hot-rolled products procurable to this specification are intended primarily for use in welded structures which are not stress relieved or heat treated to improve mechanical properties after welding. The steels are not intended for quench-and-temper heat treatment.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Form, size, and quantity of desired product (see 1.2).

(c) Chemical composition, if desired (see 1.2.3 and 3.2).

(d) Whether pickling is required (hot rolled products only).

(e) Finish, if other than that specified in 3.6.

(f) Level of packaging and packing desired (see 5.1 and 5.2).

6.3 **General information.** High-strength, low-alloy steels represent a specific type of

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steel in which enhanced mechanical properties are obtained by the addition of moderate amounts of one or more alloying elements, other than carbon. Steels of this type are normally furnished in the hot rolled or annealed condition to minimum mechanical properties. They are not intended for quenching. The user should not subject them to such treatment without assuming responsibility for the ensuing mechanical properties. Where these steels are used for fabrication by welding, no preheat or postheat is required. In certain complex structures, stress relieving may be desirable. These steels may be purchased in any of the standard forms normally available in carbon steel.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—MR
Navy—Weps
Air Force—ASD

Preparing activity:
Navy—Weps

Project No. 9515-0013

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